

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1460

Chowchilla 138 kV
May 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1460. The Interconnection Customer proposes a 63.0 MW battery energy storage facility interconnecting at the Chowchilla 138 kV substation in the Virginia area. The request was received on April 19, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 533 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,679,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LAKESIDE - FRITCH 138 kV L	MITCHELL DAM - SKAGGS ISLAND JUNCTION 2 345 kV	133.7	303	5.5
DEWALT SUBSTATION - OLD HICKORY 138 kV Line	BEAR - CLOVER 345 kV	107.8	417	7.0
BRODNAX - CHOWCHILLA 138 kV	GASTON - TECUMSEH ENERGY CTR 345 kV	107.6	268	54.2
EAGLE ROCK - BRODNAX 138 kV	TECUMSEH ENERGY CENTER - GASTON 345 kV	107.4	272	54.2
GENEVA BORDER CITY - DEWALT 138 kV Line	BEAR - CLOVER 345 kV Line	107.2	434	7.0
LAKESIDE - NATCHEZ 138 kV	MITCHELL DAM - SKAGGS ISLAND JUNCTION 2 345 kV	107.2	317	5.5
RALEIGH HILLS - EAGLE ROCK 138 kV line	TECUMSEH ENERGY CENTER - GASTON 345 kV L	107.1	278	54.2
CHANCELLOR ROW - NATCHEZ 138 kV	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 kV	106.8	298	5.5
PLANO PARKER ROAD - CHOWCHILLA 138 kV	TECUMSEH ENERGY CTR - GASTON 345 kV	106.0	260	45.8
WINDSOR DP - AVRA VALLEY 138 kV	MIRA VISTA - YARNEN 345 kV	101.1	363	5.2
WERNER WEST - WINDSOR DP 138 kV	MIRA VISTA - YARNEN 345 kV	100.4	376	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAKESIDE - FRITCH 138 kV L	Rebuild line with higher-capacity conductor	\$2,833,000	1% (\$28,000)
DEWALT SUBSTATION - OLD HICKORY 138 kV Line	Reconductor line	\$2,845,000	25% (\$718,000)
BRODNAX - CHOWCHILLA 138 kV	Reconductor line	\$478,000	40% (\$191,000)
EAGLE ROCK - BRODNAX 138 kV	Reconductor line	\$5,206,000	40% (\$2,077,000)
GENEVA BORDER CITY - DEWALT 138 kV Line	Reconductor line	\$10,386,000	25% (\$2,622,000)
LAKESIDE - NATCHEZ 138 kV	Reconductor line	\$870,000	1% (\$9,000)
RALEIGH HILLS - EAGLE ROCK 138 kV line	Reconductor line	\$1,273,000	40% (\$508,000)
CHANCELLOR ROW - NATCHEZ 138 kV	Reconductor line	\$961,000	1% (\$9,000)
CHOWCHILLA - PANO PARKER RD 138 kV	Reconductor line	\$12,304,000	36% (\$4,421,000)
WINSOR DP - AVRA VALLEY SUBSTATION 138 kV	Reconductor line	\$2,102,000	2% (\$37,000)
WERNER WEST - WINDSOR DP 138 kV	Reconductor line	\$3,317,000	2% (\$59,000)
	Total allocated to project		\$10,679,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.